

Marginal distribution

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \sim N \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} \right)$$

$$f(x_1, x_2) = \frac{1}{2\pi\sqrt{1-\rho^2}} e^{-\frac{1}{2} \cdot \frac{x_1^2 + x_2^2 - 2\rho x_1 x_2}{1-\rho^2}}$$

$$f_{x_1}(x_1) = \int_{-\infty}^{+\infty} f(x_1, x_2) dx_2$$

$$= \frac{1}{2\pi\sqrt{1-\rho^2}} \int_{-\infty}^{+\infty} e^{-\frac{1}{2} \cdot \frac{x_1^2 + x_2^2 - 2\rho x_1 x_2}{1-\rho^2}} dx_2$$

$$= \frac{1}{2\pi\sqrt{1-\rho^2}} \cdot e^{-\frac{1}{2} \cdot \frac{x_1^2}{1-\rho^2}} \int_{-\infty}^{+\infty} e^{-\frac{1}{2} \cdot \frac{x_2^2 - 2\rho x_1 x_2}{1-\rho^2}} dx_2$$

$$= \frac{1}{2\pi\sqrt{1-\rho^2}} e^{-\frac{1}{2} \cdot \frac{x_1^2}{1-\rho^2}} \int_{-\infty}^{+\infty} e^{-\frac{1}{2} \cdot \frac{x_2^2 - 2\rho x_1 x_2 + (\rho x_1)^2 - (\rho x_1)^2}{1-\rho^2}} dx_2$$

$$= \frac{1}{2\pi\sqrt{1-\rho^2}} e^{-\frac{1}{2} \cdot \frac{x_1^2}{1-\rho^2}} \times e^{\frac{1}{2} \cdot \frac{(\rho x_1)^2}{1-\rho^2}}$$

$$\times \underbrace{\sqrt{2\pi} \sqrt{1-\rho^2} \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}\sqrt{1-\rho^2}} e^{-\frac{1}{2} \cdot \frac{(x_2 - \rho x_1)^2}{1-\rho^2}} dx_2}_{\rightarrow 1}$$

$$= \frac{1}{2\pi\sqrt{1-\rho^2}} \times e^{-\frac{1}{2} \cdot \frac{x_1^2 - \rho^2 x_1^2}{1-\rho^2}} \times \sqrt{2\pi} \times \sqrt{1-\rho^2}$$

$$= \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{1}{2} \cdot \cancel{x_1^2(\cancel{1-\rho^2})}/(\cancel{1-\rho^2})} = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} x_1^2}$$

So, the marginal distribution of X_1 is $N(0,1)$

Conditional distribution:

The joint distribution is

$$f(x_1, x_2) = \frac{1}{2\pi\sqrt{1-\rho^2}} e^{-\frac{1}{2} \cdot \frac{x_1^2 + x_2^2 - 2\rho x_1 x_2}{1-\rho^2}}$$

$$f_{X_1}(x_1) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} x_1^2}$$

$$\text{So, } f_{X_2|X_1}(x_2|x_1) = \frac{f_{X_1, X_2}(x_1, x_2)}{f_{X_1}(x_1)}$$

$$= \frac{\frac{1}{2\pi\sqrt{1-\rho^2}} e^{-\frac{1}{2} \cdot \frac{x_1^2 + x_2^2 - 2\rho x_1 x_2}{1-\rho^2}}}{\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} x_1^2}}$$

$$= \frac{1}{\sqrt{2\pi}\sqrt{1-\rho^2}} e^{-\frac{1}{2} \left[\frac{x_1^2 + x_2^2 - 2\rho x_1 x_2}{1-\rho^2} - x_1^2 \right]}$$

$$= \frac{1}{\sqrt{2\pi}\sqrt{1-\rho^2}} e^{-\frac{1}{2} \frac{x_1^2 + x_2^2 - 2\rho x_1 x_2 - (1-\rho^2)x_1^2}{1-\rho^2}}$$

$$= \frac{1}{\sqrt{2\pi}\sqrt{1-\rho^2}} e^{-\frac{1}{2} \frac{\rho^2 x_1^2 + x_2^2 - 2\rho x_1 x_2}{1-\rho^2}}$$

$$= \frac{1}{\sqrt{2\pi}\sqrt{1-\rho^2}} \cdot e^{-\frac{1}{2} \frac{(x_2 - \rho x_1)^2}{1-\rho^2}}$$

$$\text{So, } x_2 | x_1 = x_1 \sim N(\rho x_1, 1-\rho^2)$$